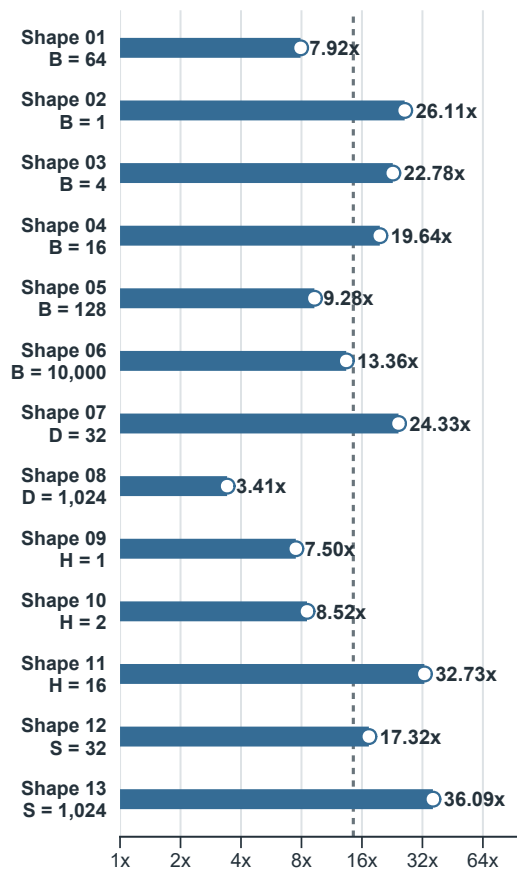


Full gains and performance retained after mechanism removal

Each cell is the fraction of complete speedup retained after replacing one mechanism family with its legal fallback

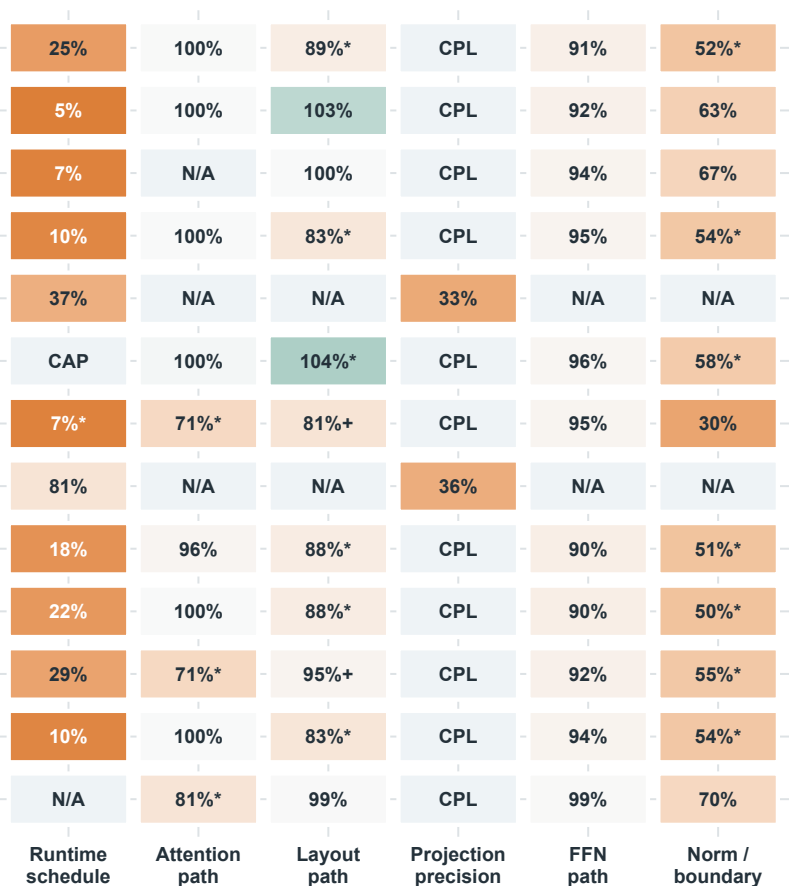
a Complete deployment

Official baseline / deployed median

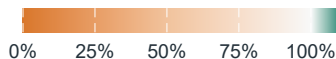


b Performance retained after removal

Ablated speedup / complete speedup; 100% = no change



Performance retained after removal



100% means no change; lower values indicate stronger dependence; above 100% means the fallback was faster.

Dashed line: 13-Shape geometric mean (14.49x). * dependency closure; + QKV-side partial.

CAP = unsafe full-batch counterfactual; CPL = active but coupled; N/A = already on fallback.

25 calls/config (Shape 06: 15); effects are non-additive.